

Task H - Feature Scaling Decision Report (Page 1 of 10)

Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
isFraud	Standardization	High skewness (5.06); Standardization is robust against extreme varia		0.035	0.1838
TransactionAmt	Standardization	High skewness (14.37); Standardization is robust against extreme var		135.0268	239.1626
card1	Standardization	Continuous numerical feature; Standardization selected to center data		9898.7646	4901.1605
card2	Standardization	Continuous numerical feature; Standardization selected to center data		362.5556	156.5954
card3	Standardization	High skewness (2.02); Standardization is robust against extreme varia		153.1865	11.3226
card5	Standardization	Continuous numerical feature; Standardization selected to center data		199.4716	41.1577
card6	Standardization	Continuous numerical feature; Standardization selected to center data		0.7477	0.4343
addr1	Standardization	Continuous numerical feature; Standardization selected to center data		290.7341	95.9142
addr2	Standardization	High skewness (-15.39); Standardization is robust against extreme va		86.8228	2.5373
dist1	Standardization	High skewness (8.23); Standardization is robust against extreme varia		52.5839	242.3519
P_emaildomain	Standardization	High skewness (3.91); Standardization is robust against extreme varia		1.8335	3.7573
C1	Standardization	High skewness (23.96); Standardization is robust against extreme var		14.0923	133.5693
C3	Standardization	High skewness (88.95); Standardization is robust against extreme var		0.0056	0.1505
C5	Standardization	High skewness (5.79); Standardization is robust against extreme varia		5.5713	25.7866
C7	Standardization	High skewness (27.19); Standardization is robust against extreme var		2.8485	61.7275
C9	Standardization	High skewness (5.69); Standardization is robust against extreme varia		4.4801	16.6746
C13	Standardization	High skewness (8.99); Standardization is robust against extreme varia		32.5393	129.364
D1	Standardization	Continuous numerical feature; Standardization selected to center data		94.1512	157.5481
D2	Standardization	Continuous numerical feature; Standardization selected to center data		135.0602	133.4326
D3	Standardization	High skewness (6.23); Standardization is robust against extreme varia		19.2875	47.5565

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Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
D4	Standardization	Continuous numerical feature; Standardization selected to center data		107.3927	169.4888
D5	Standardization	High skewness (5.20); Standardization is robust against extreme varia		25.3701	63.4496
D10	Standardization	Continuous numerical feature; Standardization selected to center data		109.9524	174.3201
D11	Standardization	High skewness (2.06); Standardization is robust against extreme varia		97.6152	144.6349
D15	Standardization	Continuous numerical feature; Standardization selected to center data		163.7445	186.806
M2	Standardization	High skewness (3.80); Standardization is robust against extreme varia		0.0575	0.2328
M3	Standardization	High skewness (2.42); Standardization is robust against extreme varia		0.1147	0.3186
M5	Standardization	Continuous numerical feature; Standardization selected to center data		0.1821	0.386
M6	Standardization	Continuous numerical feature; Standardization selected to center data		0.6726	0.4693
M7	Standardization	High skewness (3.87); Standardization is robust against extreme varia		0.0557	0.2294
M8	Standardization	Continuous numerical feature; Standardization selected to center data		0.1508	0.3578
M9	Standardization	High skewness (3.52); Standardization is robust against extreme varia		0.0654	0.2473
V2	Standardization	High skewness (9.44); Standardization is robust against extreme varia		1.0238	0.1758
V3	Standardization	High skewness (7.56); Standardization is robust against extreme varia		1.0412	0.2362
V4	Standardization	Continuous numerical feature; Standardization selected to center data		0.8465	0.3195
V5	Standardization	Continuous numerical feature; Standardization selected to center data		0.877	0.3455
V6	Standardization	High skewness (9.37); Standardization is robust against extreme varia		1.0241	0.1753
V7	Standardization	High skewness (7.73); Standardization is robust against extreme varia		1.0384	0.2242
V9	Standardization	High skewness (9.67); Standardization is robust against extreme varia		1.0219	0.1656
V10	Standardization	Continuous numerical feature; Standardization selected to center data		0.4639	0.3786

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Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
V12	Standardization	Continuous numerical feature; Standardization selected to center data		0.5597	0.4765
V13	Standardization	Continuous numerical feature; Standardization selected to center data		0.5992	0.4967
V15	Standardization	High skewness (2.84); Standardization is robust against extreme varia		0.1066	0.313
V16	Standardization	High skewness (4.15); Standardization is robust against extreme varia		0.1076	0.3227
V17	Standardization	High skewness (4.10); Standardization is robust against extreme varia		0.1168	0.3431
V18	Standardization	High skewness (4.33); Standardization is robust against extreme varia		0.1179	0.3499
V19	Standardization	Continuous numerical feature; Standardization selected to center data		0.8164	0.3972
V20	Standardization	Continuous numerical feature; Standardization selected to center data		0.8478	0.4288
V21	Standardization	High skewness (2.55); Standardization is robust against extreme varia		0.113	0.3194
V22	Standardization	High skewness (3.94); Standardization is robust against extreme varia		0.1153	0.3384
V23	Standardization	High skewness (12.99); Standardization is robust against extreme var		1.0303	0.2315
V24	Standardization	High skewness (9.74); Standardization is robust against extreme varia		1.0506	0.2858
V25	Standardization	High skewness (-2.88); Standardization is robust against extreme vari		0.9805	0.1731
V26	Standardization	High skewness (2.85); Standardization is robust against extreme varia		0.9896	0.1954
V29	Standardization	Continuous numerical feature; Standardization selected to center data		0.3878	0.4766
V30	Standardization	Continuous numerical feature; Standardization selected to center data		0.3541	0.535
V31	Standardization	High skewness (2.69); Standardization is robust against extreme varia		0.1226	0.3354
V32	Standardization	High skewness (3.83); Standardization is robust against extreme varia		0.1241	0.347
V33	Standardization	High skewness (2.58); Standardization is robust against extreme varia		0.1139	0.3212
V34	Standardization	High skewness (3.22); Standardization is robust against extreme varia		0.1212	0.337

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Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
V35	Standardization	Continuous numerical feature; Standardization selected to center data		0.5426	0.436
V36	Standardization	Continuous numerical feature; Standardization selected to center data		0.5792	0.4554
V37	Standardization	High skewness (26.66); Standardization is robust against extreme var		1.0771	0.5855
V38	Standardization	High skewness (21.59); Standardization is robust against extreme var		1.1159	0.7294
V39	Standardization	High skewness (7.00); Standardization is robust against extreme varia		0.1186	0.3892
V40	Standardization	High skewness (7.80); Standardization is robust against extreme varia		0.1265	0.4348
V42	Standardization	High skewness (3.07); Standardization is robust against extreme varia		0.1114	0.3311
V43	Standardization	High skewness (3.80); Standardization is robust against extreme varia		0.1206	0.374
V44	Standardization	High skewness (34.25); Standardization is robust against extreme var		1.0599	0.5413
V45	Standardization	High skewness (27.33); Standardization is robust against extreme var		1.0862	0.619
V47	Standardization	High skewness (11.91); Standardization is robust against extreme var		1.0275	0.1967
V48	Standardization	Continuous numerical feature; Standardization selected to center data		0.3832	0.4294
V49	Standardization	Continuous numerical feature; Standardization selected to center data		0.2839	0.4925
V50	Standardization	High skewness (2.49); Standardization is robust against extreme varia		0.1176	0.3246
V51	Standardization	High skewness (3.04); Standardization is robust against extreme varia		0.1218	0.3499
V52	Standardization	High skewness (4.01); Standardization is robust against extreme varia		0.1304	0.38
V53	Standardization	Continuous numerical feature; Standardization selected to center data		0.5776	0.477
V54	Standardization	Continuous numerical feature; Standardization selected to center data		0.62	0.4985
V55	Standardization	High skewness (11.18); Standardization is robust against extreme var		1.0588	0.3656
V56	Standardization	High skewness (25.17); Standardization is robust against extreme var		1.1052	0.6178

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Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
V57	Standardization	High skewness (2.94); Standardization is robust against extreme varia		0.1116	0.3284
V58	Standardization	High skewness (4.11); Standardization is robust against extreme varia		0.1152	0.3506
V59	Standardization	High skewness (4.72); Standardization is robust against extreme varia		0.1169	0.3566
V60	Standardization	High skewness (5.37); Standardization is robust against extreme varia		0.1239	0.3929
V61	Standardization	Continuous numerical feature; Standardization selected to center data		0.8298	0.4071
V62	Standardization	Continuous numerical feature; Standardization selected to center data		0.8676	0.4511
V63	Standardization	High skewness (3.10); Standardization is robust against extreme varia		0.1137	0.3342
V64	Standardization	High skewness (4.17); Standardization is robust against extreme varia		0.1233	0.3817
V66	Standardization	Continuous numerical feature; Standardization selected to center data		0.981	0.2016
V67	Standardization	Continuous numerical feature; Standardization selected to center data		0.9984	0.2293
V69	Standardization	Continuous numerical feature; Standardization selected to center data		0.3902	0.479
V70	Standardization	Continuous numerical feature; Standardization selected to center data		0.3547	0.535
V71	Standardization	High skewness (2.82); Standardization is robust against extreme varia		0.1223	0.343
V72	Standardization	High skewness (3.92); Standardization is robust against extreme varia		0.1262	0.3662
V73	Standardization	High skewness (2.91); Standardization is robust against extreme varia		0.1217	0.3455
V74	Standardization	High skewness (3.08); Standardization is robust against extreme varia		0.1323	0.3703
V75	Standardization	Continuous numerical feature; Standardization selected to center data		0.5443	0.4739
V76	Standardization	Continuous numerical feature; Standardization selected to center data		0.5876	0.4959
V77	Standardization	High skewness (14.35); Standardization is robust against extreme var		1.0738	0.4921
V78	Standardization	High skewness (16.53); Standardization is robust against extreme var		1.1227	0.7219

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Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
V79	Standardization	High skewness (4.26); Standardization is robust against extreme varia		0.1162	0.3535
V80	Standardization	High skewness (6.17); Standardization is robust against extreme varia		0.1222	0.3812
V81	Standardization	High skewness (6.44); Standardization is robust against extreme varia		0.1296	0.4203
V82	Standardization	Continuous numerical feature; Standardization selected to center data		0.8446	0.3893
V83	Standardization	Continuous numerical feature; Standardization selected to center data		0.882	0.4338
V84	Standardization	High skewness (3.03); Standardization is robust against extreme varia		0.1164	0.337
V85	Standardization	High skewness (4.11); Standardization is robust against extreme varia		0.1272	0.3902
V86	Standardization	High skewness (18.35); Standardization is robust against extreme var		1.0551	0.3875
V87	Standardization	High skewness (16.02); Standardization is robust against extreme var		1.0844	0.4724
V90	Standardization	Continuous numerical feature; Standardization selected to center data		0.4019	0.4756
V91	Standardization	Continuous numerical feature; Standardization selected to center data		0.357	0.5392
V92	Standardization	High skewness (2.83); Standardization is robust against extreme varia		0.1276	0.3503
V93	Standardization	High skewness (3.94); Standardization is robust against extreme varia		0.1314	0.3748
V94	Standardization	High skewness (2.39); Standardization is robust against extreme varia		0.1163	0.3206
V95	Standardization	High skewness (30.25); Standardization is robust against extreme var		1.0375	21.0288
V96	Standardization	High skewness (26.04); Standardization is robust against extreme var		3.0036	40.2289
V97	Standardization	High skewness (25.33); Standardization is robust against extreme var		1.718	27.6932
V98	Standardization	High skewness (6.39); Standardization is robust against extreme varia		0.062	0.2849
V99	Standardization	High skewness (11.07); Standardization is robust against extreme var		0.8945	2.7219
V100	Standardization	High skewness (8.74); Standardization is robust against extreme varia		0.2734	0.9469

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Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
V101	Standardization	High skewness (30.64); Standardization is robust against extreme var		0.8888	20.5772
V102	Standardization	High skewness (27.00); Standardization is robust against extreme var		1.8263	35.919
V103	Standardization	High skewness (26.04); Standardization is robust against extreme var		1.2786	25.6789
V104	Standardization	High skewness (14.89); Standardization is robust against extreme var		0.0854	0.6485
V105	Standardization	High skewness (20.90); Standardization is robust against extreme var		0.281	3.3731
V106	Standardization	High skewness (19.49); Standardization is robust against extreme var		0.1645	1.8252
V115	Standardization	High skewness (6.59); Standardization is robust against extreme varia		1.0324	0.1903
V123	Standardization	High skewness (12.88); Standardization is robust against extreme var		1.0311	0.2281
V124	Standardization	High skewness (6.09); Standardization is robust against extreme varia		1.0929	0.374
V125	Standardization	High skewness (9.00); Standardization is robust against extreme varia		1.0504	0.28
V126	Standardization	High skewness (29.59); Standardization is robust against extreme var		129.9107	2346.3355
V127	Standardization	High skewness (25.69); Standardization is robust against extreme var		336.4312	4237.5574
V128	Standardization	High skewness (25.06); Standardization is robust against extreme var		203.9858	3009.4696
V129	Standardization	High skewness (240.34); Standardization is robust against extreme va		8.7643	113.803
V130	Standardization	High skewness (26.30); Standardization is robust against extreme var		92.1161	315.8837
V131	Standardization	High skewness (89.62); Standardization is robust against extreme var		31.1167	161.1204
V132	Standardization	High skewness (30.24); Standardization is robust against extreme var		103.4584	2265.5106
V133	Standardization	High skewness (26.87); Standardization is robust against extreme var		204.7804	3795.3199
V134	Standardization	High skewness (25.99); Standardization is robust against extreme var		145.8949	2772.2586
V135	Standardization	High skewness (144.92); Standardization is robust against extreme va		17.241	293.7704

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Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
V136	Standardization	High skewness (49.03); Standardization is robust against extreme var		38.7998	451.6897
V137	Standardization	High skewness (90.61); Standardization is robust against extreme var		26.3512	348.2415
V279	Standardization	High skewness (30.24); Standardization is robust against extreme var		1.123	21.0218
V280	Standardization	High skewness (24.78); Standardization is robust against extreme var		1.967	27.8516
V281	Standardization	High skewness (10.02); Standardization is robust against extreme var		0.0876	0.5122
V282	Standardization	High skewness (3.83); Standardization is robust against extreme varia		0.8176	0.9209
V283	Standardization	High skewness (12.41); Standardization is robust against extreme var		0.9911	1.5571
V284	Standardization	High skewness (5.30); Standardization is robust against extreme varia		0.0885	0.3384
V285	Standardization	High skewness (12.29); Standardization is robust against extreme var		1.1676	3.2824
V286	Standardization	High skewness (7.59); Standardization is robust against extreme varia		0.0315	0.1909
V287	Standardization	High skewness (8.84); Standardization is robust against extreme varia		0.3586	1.079
V288	Standardization	High skewness (2.56); Standardization is robust against extreme varia		0.184	0.4306
V289	Standardization	High skewness (3.51); Standardization is robust against extreme varia		0.2355	0.5987
V290	Standardization	High skewness (34.57); Standardization is robust against extreme var		1.103	0.7689
V291	Standardization	High skewness (50.16); Standardization is robust against extreme var		1.6598	16.2524
V292	Standardization	High skewness (51.23); Standardization is robust against extreme var		1.2399	3.775
V293	Standardization	High skewness (30.58); Standardization is robust against extreme var		0.9426	20.5887
V294	Standardization	High skewness (23.85); Standardization is robust against extreme var		2.3138	39.5262
V295	Standardization	High skewness (25.34); Standardization is robust against extreme var		1.4334	25.9627
V296	Standardization	High skewness (18.95); Standardization is robust against extreme var		0.3282	3.2613

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Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
V297	Standardization	High skewness (13.28); Standardization is robust against extreme var		0.089	0.6283
V298	Standardization	High skewness (19.89); Standardization is robust against extreme var		0.2988	3.175
V299	Standardization	High skewness (18.68); Standardization is robust against extreme var		0.1717	1.7242
V300	Standardization	High skewness (10.86); Standardization is robust against extreme var		0.0454	0.2893
V301	Standardization	High skewness (10.55); Standardization is robust against extreme var		0.0519	0.318
V302	Standardization	High skewness (2.80); Standardization is robust against extreme varia		0.2518	0.4819
V303	Standardization	High skewness (5.16); Standardization is robust against extreme varia		0.2831	0.6236
V304	Standardization	High skewness (3.29); Standardization is robust against extreme varia		0.2642	0.5282
V306	Standardization	High skewness (29.27); Standardization is robust against extreme var		139.7463	2348.8318
V307	Standardization	High skewness (23.67); Standardization is robust against extreme var		408.6725	4391.9593
V308	Standardization	High skewness (24.53); Standardization is robust against extreme var		230.409	3021.9014
V309	Standardization	High skewness (224.88); Standardization is robust against extreme va		10.9958	116.2534
V310	Standardization	High skewness (19.50); Standardization is robust against extreme var		118.1924	352.9797
V311	Standardization	High skewness (323.83); Standardization is robust against extreme va		4.2021	102.3742
V312	Standardization	High skewness (73.60); Standardization is robust against extreme var		39.1731	172.1271
V313	Standardization	High skewness (14.04); Standardization is robust against extreme var		21.3057	95.8052
V314	Standardization	High skewness (10.76); Standardization is robust against extreme var		43.2263	173.4444
V315	Standardization	High skewness (12.13); Standardization is robust against extreme var		26.7495	116.7345
V316	Standardization	High skewness (30.10); Standardization is robust against extreme var		109.8166	2270.0159
V317	Standardization	High skewness (24.60); Standardization is robust against extreme var		247.6021	3980.0126

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Feature	Chosen Method	Reasoning		Param 1 (Min / Mean)	Param 2 (Max / Std)
V318	Standardization	High skewness (25.48); Standardization is robust against extreme var		162.1504	2793.3224
V319	Standardization	High skewness (181.84); Standardization is robust against extreme va		18.3722	332.3023
V320	Standardization	High skewness (70.52); Standardization is robust against extreme var		42.0712	473.4948
V321	Standardization	High skewness (123.56); Standardization is robust against extreme va		28.3262	382.0503
ProductCD_W	Standardization	Continuous numerical feature; Standardization selected to center data		0.7445	0.4361
ProductCD_H	Standardization	High skewness (3.87); Standardization is robust against extreme varia		0.0559	0.2298
ProductCD_C	Standardization	High skewness (2.40); Standardization is robust against extreme varia		0.116	0.3203
ProductCD_R	Standardization	High skewness (3.57); Standardization is robust against extreme varia		0.0638	0.2445
card4_mastercard	Standardization	Continuous numerical feature; Standardization selected to center data		0.3204	0.4666
card4_visa	Standardization	Continuous numerical feature; Standardization selected to center data		0.6542	0.4756
M4_M2	Standardization	High skewness (2.64); Standardization is robust against extreme varia		0.1014	0.3018
M4_M0	Standardization	Continuous numerical feature; Standardization selected to center data		0.8092	0.393
M4_M1	Standardization	High skewness (2.88); Standardization is robust against extreme varia		0.0895	0.2854